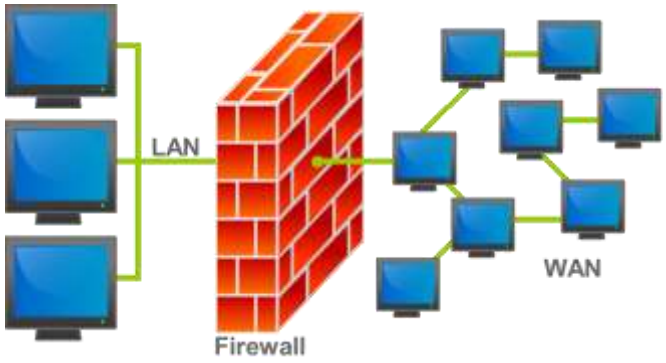
PRESENTATION ON FIREWALL

BY

MOHAMMED MAAJIDH

WHAT IS FIREWALL ?

- A firewall can either be software-based or hardware-based and is used to help keep a network secure. A system designed to prevent unauthorized access to or from a private network. Firewalls can be implemented in both hardware and software, or a combination of both



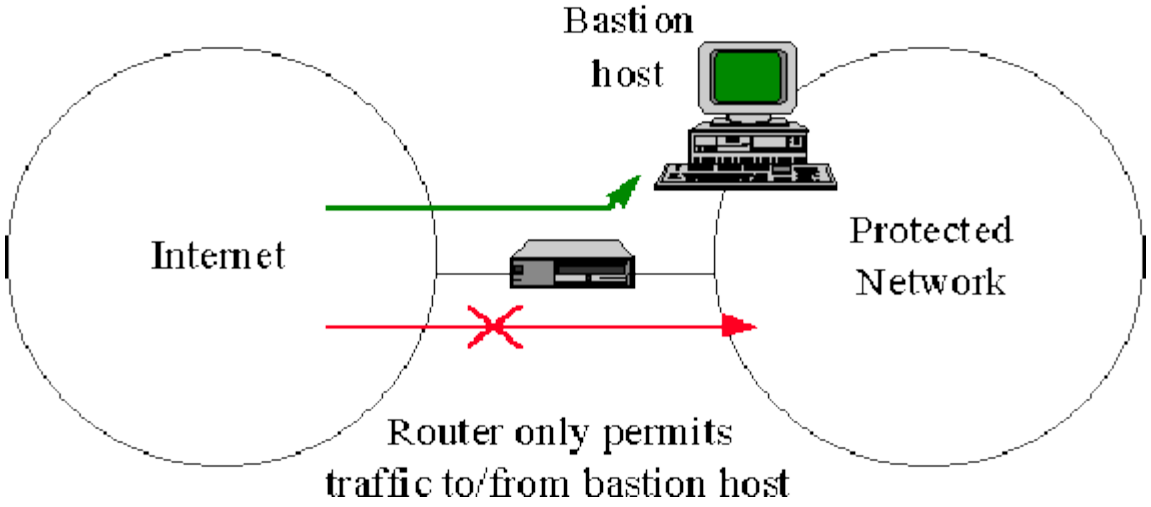
TYPES OF FIREWALL
THEY MAY BE
1)HARDWARE FIREWALL
2)SOFTWARE FIREWALL

- Network layer firewall
- Application layer firewall
- Circuit layer firewall
- Stateful multi-layer inspection firewall
- Proxy firewall
- Host-based firewall
- Packet filtering
- Hybrid firewall

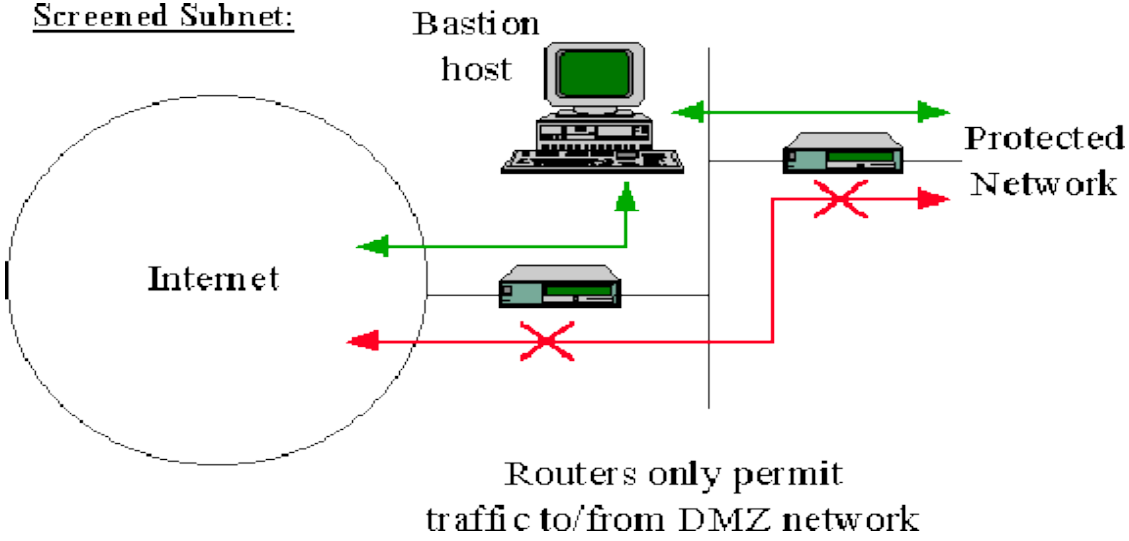
NETWORK LAYER FIREWALL

- The first generation of firewalls (c. 1988) worked at the network level by inspecting packet headers and filtering traffic based on the IP address of the source and the destination, the port and the service. Some of these primeval security applications could also filter packets based on protocols, the domain name of the source and a few other attributes.
- One important difference about many network layer firewalls is that they route traffic directly through them, which means in order to use one, you either need to have a validly-assigned IP address block or a private Internet address block

Screened Host Firewall:



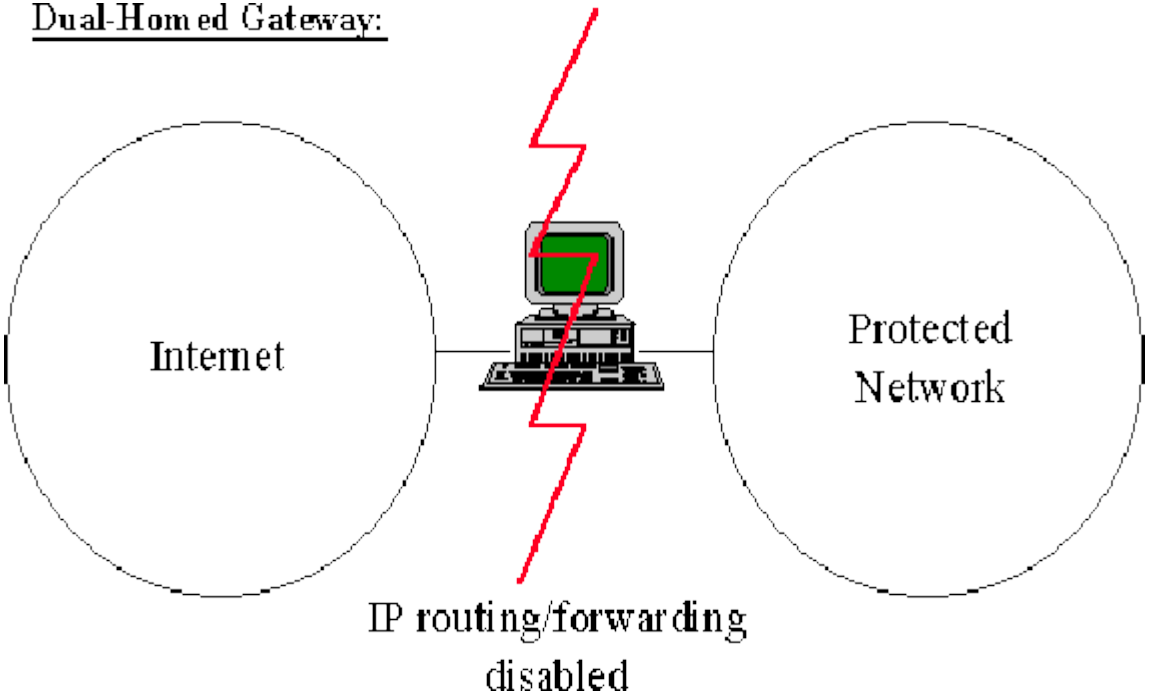
Screened Subnet:



APPLICATION LAYER FIREWALL

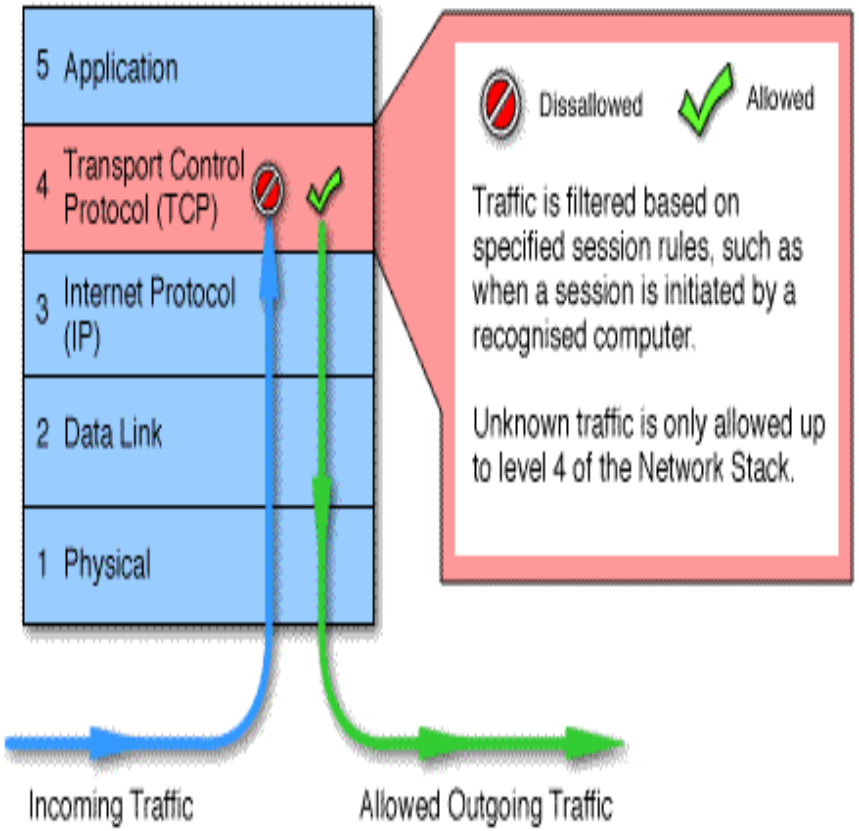
- Application-level firewalls (sometimes called proxies) have been looking more deeply into the application data going through their filters. Application layer firewalls defined, are hosts running proxy servers, which permit no traffic directly between networks, and they perform elaborate logging and examination of traffic passing through them
- Application-level filtering may include protection against spam and viruses as well, and be able to block undesirable Web sites based on content rather than just their IP address.

Dual-Homed Gateway:



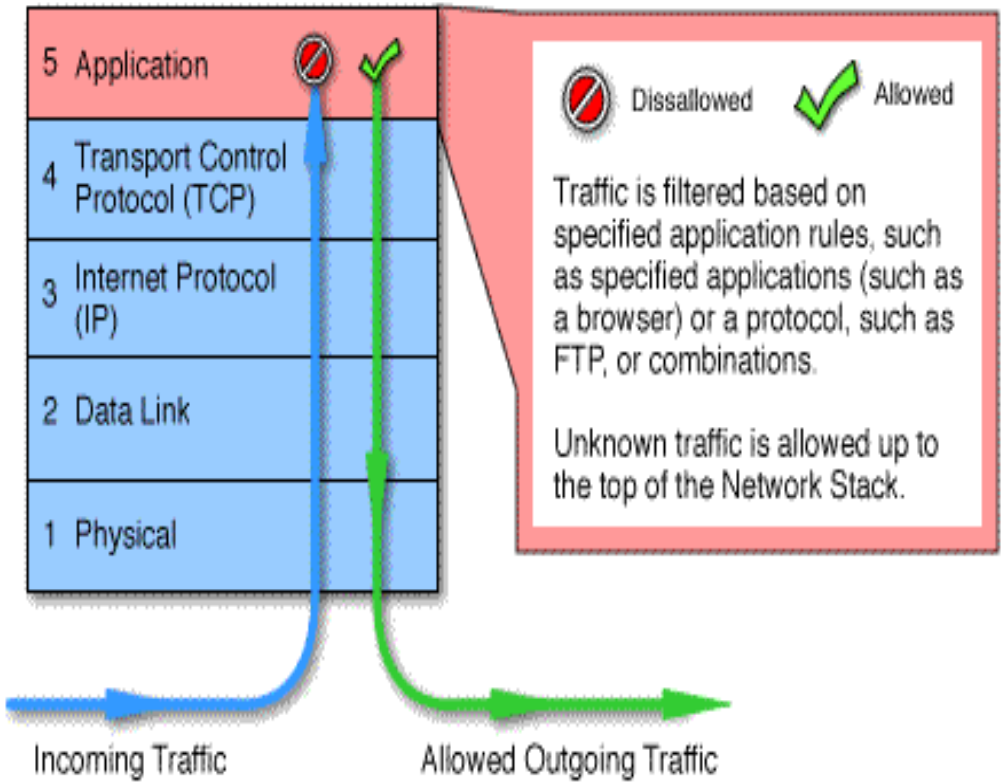
CIRCUIT-LEVEL FIREWALL

- These applications, which represent the second-generation of firewall technology, monitor TCP handshaking between packets to make sure a session is legitimate. Traffic is filtered based on specified session rules and may be restricted to recognized computers only. Circuit-level firewalls hide the network itself from the outside, which is useful for denying access to intruders. But they don't filter individual packets. Applies security mechanism when a TCP or UDP connection is established. Once the connection has been made, packets can flow between the hosts without further checking. Circuit gateways firewalls function at the network transport layer. They allow or deny connections based on addresses and prevent direct connection between networks


5 Application
4 Transport Control Protocol (TCP)
3 Internet Protocol (IP)
2 Data Link
1 Physical
Dissallowed
Allowed
Traffic is filtered based on specified session rules, such as when a session is initiated by a recognised computer.
Unknown traffic is only allowed up to level 4 of the Network Stack.
Incoming Traffic
Allowed Outgoing Traffic

STATEFUL MULTI-LAYER INSPECTION FIREWALL

- Stateful multilayer inspection firewalls combine the aspects of the other three types of firewalls. SML vendors claim that their products deploy the best features of the other three firewall types. They filter packets at the network level and they recognize and process application-level data, but since they don't employ proxies, they deliver reasonably good performance in spite of the deep packet analysis


5 Application
4 Transport Control Protocol (TCP)
3 Internet Protocol (IP)
2 Data Link
1 Physical
Dissallowed
Allowed
Traffic is filtered based on specified application rules, such as specified applications (such as a browser) or a protocol, such as FTP, or combinations.
Unknown traffic is allowed up to the top of the Network Stack.
Incoming Traffic
Allowed Outgoing Traffic

PROXY FIREWALL

- Proxy firewalls offer more security than other types of firewalls, but this is at the expense of speed and functionality, as they can limit which applications your network can support.
- Proxy firewalls also provide comprehensive, protocol-aware security analysis for the protocols they support.
- This allows them to make better security decisions than products that focus purely on packet header information.
- Intercepts all messages entering and leaving the network.
- The proxy server efficiently hides the true network addresses.

HOST BASED FIREWALL

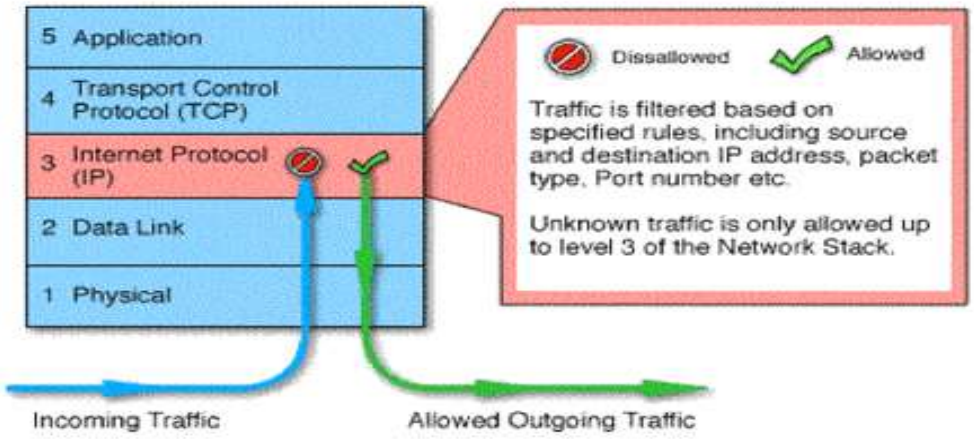
- Network perimeter firewalls cannot provide protection for traffic generated inside a trusted network. For this reason, host-based firewalls running on individual computers are needed. Host-based firewalls, of which Windows Firewall with Advanced Security is an example, protect a host from unauthorized access and attack.

PACKET FILTERING

- Firewalls fall into four broad categories: packet filters, circuit level gateways, application level gateways and stateful multilayer inspection firewalls.
- Packet filtering firewalls work at the network level of the OSI model, or the IP layer of TCP/IP
- They are usually part of a router. A router is a device that receives packets from one network and forwards them to another network. In a packet filtering firewall each packet is compared to a set of criteria before it is forwarded

FILTERING FIREWALLS CLASSIFICATION

- **Static Filtering** – is being implemented by most routers. Rules of filters are adjusted manually.
- **Dynamic Filtering** – allows filtering rules to change depending on responses to outside processes.



HYBRID FIREWALL

- Hybrid firewalls as the name suggests, represent a combination of technologies. A hybrid firewall may consist of a pocket filtering combined with an application proxy firewall, or a circuit gateway combined with an application proxy firewall.

LIST OF FIREWALLS ON OS

List of firewall using in Linux operating system?

- Lptables
- Lpcop
- Shorewall
- UFW – Uncomplicated Firewall
- OpenBSD and PF
- EBox platform
- Monowall
- Clear os
- pfSense
- Smoothwall Advanced

List of firewall using in Windows operating system?

- Zone Alarm firewall
- Shardaccess
- Mpssvc

THANK YOU